

NANYANG TECHNOLOGICAL UNIVERSITY
School of Social Sciences
HE3002 Macroeconomics III

PROBLEM SET 2
The Goods Market in an Open Economy

1. Consider an open economy characterised by the equations below:

$$C = c_0 + c_1(Y - T)$$

$$I = d_0 + d_1Y$$

$$IM = m_1Y$$

$$X = x_1Y^*$$

The parameters m_1 and x_1 are the propensities to import and export. Assume that the real exchange rate is fixed at a value of 1 and treat foreign income, Y^* , as fixed. Also assume that taxes are fixed and that government purchases are exogenous (i.e. decided by the government). We explore the effectiveness of changes in G under alternative assumptions about the propensity to import.

- (a) Write the equilibrium condition in the market for domestic goods and solve for Y .
 - (b) Suppose government purchases increase by one unit. What is the effect on output? (Assume that $0 < m_1 < c_1 + m_1 < 1$. Explain why.)
 - (c) How do net exports change when government purchases increase by one unit?
2. Refer to Question 1. Now consider two economies, one with $m_1=0.5$ and the other with $m_1=0.1$. Each economy is characterised by $(c_1 + d_1) = 0.6$.
- (a) Suppose one of the economies is much larger than the other. Which economy do you expect to have the larger value of m_1 ? Explain.
 - (b) Calculate your answers to parts (b) and (c) for each economy by substituting the appropriate parameter values.
 - (c) In which economy will fiscal policy have a larger effect on output? In which economy will fiscal policy have a larger effect on net exports?
3. Consider an open economy in which the real exchange rate is fixed and equal to one. Consumption, investment, government spending, and taxes are given by:

$$C = 10 + 0.8(Y - T)$$

$$I = 10$$

$$G = 10$$

$$T = 10$$

$$IM = 0.3Y$$

$$X = 0.3Y^*$$

- (a) Solve for equilibrium output in the domestic economy, given Y^* . What is the multiplier in this economy? If we were to close the economy – so exports and imports were identically equal to zero – what would the multiplier be? Why would the multiplier be different in a closed economy?
- (b) Assume that the foreign economy is characterised by the same equations as the domestic economy (with asterisks reversed). Use the two sets of equations to solve for the equilibrium output of each country. [Hint: use the equations for the foreign economy to solve for Y^* as a function of Y and substitute this solution for Y^* in part (a).] What is the multiplier for each country now? Why is it different from the open economy multiplier in part (a)?
- (c) Assume that the domestic government, G , has a target level of output of 125. Assuming that the foreign government does not change G^* , what is the increase in G necessary to achieve the target output in the domestic economy? Solve for net exports and the budget deficit in each country.
- (d) Suppose each government has a target level of output of 125 and that each government increases government spending by the same amount. What is the common increase in G and G^* necessary to achieve the target output in both countries? Solve for net exports and the budget deficit in each country.
- (e) Why is fiscal coordination, such as the common increase in G and G^* in part (d), difficult to achieve in practice?